

Homework for week 7

1. *Reduction of order.* Kreyszig 2.1, Problems 4, 6, 8
2. *2nd order initial value problems:* Kreyszig 2.1, Problems 15, 16, 19
3. *Linear homogeneous constant coefficient ODE-s.*
Kreyszig 2.2, Problems 1, 7, 11, 12, 23, 27, 30
4. *Linear independence:* Verify that the functions

$$y_1 = \tanh(x), \quad y_2 = 1 - \frac{2}{e^{2x} + 1}$$

are particular solutions of the differential equation

$$y'' + 2yy' = 0.$$

Check whether they are linearly independent by calculating the Wronskian

$$W(x) = y_1 y_2' - y_2 y_1'.$$

Evaluate $W(x)$ for an x where it is easy to compute.